The status of academic research on salmon farming: a living evidence map

Results

Review process

The review process is outlined in the flow diagram in Figure 1. Across the 7 databases searched, a total of 118,527 search results was obtained. Following deduplication, 47,094 unique candidate records remained. Title and abstract repair and enrichment supplemented 2,218 records, whilst a sweep for retractions identified 12 records that were excluded from further analysis.

Following screening of titles and abstracts, 21,493 were included in the map, and passed on to annotation. Species annotations were obtained for all 21,493 records, geography for 8,923 records, and topic classifications for 21,246 records.

Figure 1. *Flow of records through database searching, deduplication, record repair and enrichment, retraction screening, title-and-abstract screening, annotation and inclusion in the Living Evidence Map.*

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Database provenance was retained through deduplication, allowing the contribution and overlap of the 7 bibliographic sources to be compared for the deduplicated corpus and the definitive final included corpus. Figure 2 shows records contributed uniquely by each database, while Figure 3 shows only multi-database overlaps.

Figure 2. *Unique contribution of each bibliographic database. Records are counted as unique when they were retrieved from that database and no other searched database. Black bars show unique records in the deduplicated corpus; olive-green bars show the subset retained in the definitive final included corpus.*

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Figure 3. *Overlap among bibliographic databases. Only records retrieved from two or more databases are shown; single-database records are presented in Figure 2. Black bars show exact database intersections in the deduplicated corpus and olive-green bars show the corresponding records retained in the definitive final included corpus. The matrix indicates the databases represented in each intersection.*

Figure 3. *Overlap among bibliographic databases. Only records retrieved from two or more databases are shown; single-database records are presented in Figure 2. Black bars show exact database intersections in the deduplicated corpus and olive-green bars show the corresponding records retained in the definitive final included corpus. The matrix indicates the databases represented in each intersection.*

Publication year

The volume of academic evidence on salmon farming increased markedly over time. There were 1,189 included records published in 2025, the highest complete publication year in the current corpus, compared with just 516 in 2010.

The three largest focal species categories were Rainbow trout (8,960 records; 41.7%), Atlantic salmon (8,072; 37.6%) and Unspecified species (4,489; 20.9%). Species categories are not mutually exclusive because individual records can refer to more than one eligible farmed species.

Figure 4. *Records by publication year and focal farmed species.*

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The thematic composition of the literature also expanded through time. The largest high-level topic was Production, followed by Environment. Because records can receive more than one high-level topic, annual stacked totals represent topic-record assignments rather than mutually exclusive studies.

Figure 5. *Topic-assignment frequency for high-level topics by publication year.*

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Geography

A substantive study country could be resolved for 8,923 records (41.5%), while 12,570 included records (58.5%) did not contain sufficient evidence for a defensible country-level assignment. Among records with assigned

geography (Figure 6), the largest country evidence bases were Norway (1,989 records), Canada (1,152), United Kingdom (1,050), Chile (1,028), United States (669), Turkey (493). Countries demonstrated strong species-specificity, as expected (Figure 7), with the major Atlantic salmon producing countries contributing most to the evidence base, and a large number of other countries contributing collectively to rainbow trout research (e.g. Turkey). Records may be assigned to more than one country where the substantive study spans multiple countries.

Figure 6. *Global distribution of included records by substantive study country.*

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Figure 7. *Records per country for the 20 most frequently represented study countries, by focal farmed species.*

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Topics

At least one topic was assigned to 21,246 of the 21,493 included records; 247 records (1.1%) remained uncoded for topic. Across coded records, the final ontology contained 60,221 distinct record-topic assignments.

The evidence base was strongly oriented towards Production, represented in 15,820 records (73.6%). Environment was the second-largest theme (5,185; 24.1%), followed by Industry and governance (2,745; 12.8%), Methods (2,684; 12.5%) and Product (2,594; 12.1%). People and society (1,046; 4.9%) and Inputs and resources (835; 3.9%) were substantially smaller. Theme percentages are not mutually exclusive because a record can be assigned to more than one theme.

Figure 8. *High-level topic frequency by focal farmed species. Numbers at bar ends are unique records assigned to each high-level theme; stacked species segments may sum above the unique-record total for multi-species records.*

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Production

Production research was centred on fish performance and biology, fish health, feed and nutrition, breeding and reproduction, and stress and mortality. The most frequently assigned terminal topic was Physiology, metabolism and biological responses (5,002 records; 23.3%), followed by

Growth and development (3,126; 14.5%). Other major concentrations included Other diseases - Prevention and treatment (2,004; 9.3%), Other diseases - Epidemiology (1,658; 7.7%), Other diseases - General (1,504; 7.0%) and Alternative feed ingredients (1,482; 6.9%). Sea lice remained a substantial research cluster, including 770 records on Sea lice control and treatment and 576 on Sea lice epidemiology.

Figure 9. *Topic-assignment frequency within the Production theme.*

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Environment

Environmental research was concentrated in Physiology, metabolism and biological responses (5,002; 23.3%), Growth and development (3,126; 14.5%), Other diseases - Prevention and treatment (2,004; 9.3%) and Other diseases - Epidemiology (1,658; 7.7%).

Figure 10. *Topic-assignment frequency within the Environment theme.*

Figure 10. *Topic-assignment frequency within the Environment theme.*

Methods

Methods research was dominated by Physiology, metabolism and biological responses (5,002; 23.3%), followed by Growth and development (3,126; 14.5%), Other diseases - Prevention and treatment (2,004; 9.3%) and Other diseases - Epidemiology (1,658; 7.7%).

Figure 11. *Topic-assignment frequency within the Methods theme.*

Figure 11. *Topic-assignment frequency within the Methods theme.*

Industry and governance

Within Industry and governance, Physiology, metabolism and biological responses was the largest terminal topic (5,002; 23.3%), followed by Growth and development (3,126; 14.5%), Other diseases - Prevention and treatment (2,004; 9.3%), Other diseases - Epidemiology (1,658; 7.7%) and Other diseases - General (1,504; 7.0%).

Figure 12. *Topic-assignment frequency within the Industry and governance theme.*

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Product

Product research was led by Physiology, metabolism and biological responses (5,002; 23.3%), Growth and development (3,126; 14.5%) and Other diseases - Prevention and treatment (2,004; 9.3%). Other diseases - Epidemiology and Other diseases - General were also prominent.

Figure 13. *Topic-assignment frequency within the Product theme.*

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People and society

People and society was much smaller than the Production and Environment evidence bases. The largest terminal topics were Physiology, metabolism and biological responses (5,002; 23.3%), Growth and development (3,126; 14.5%) and Other diseases - Prevention and treatment (2,004; 9.3%). Topics relating to Employment and job creation, Livelihoods and income, Indigenous peoples and rights, Food availability and nutritional security, Culture, identity and traditions, Working conditions and occupational health, Distribution of benefits and harms, Local and Indigenous knowledge, Displacement, disruption and wellbeing, Multiple or general social impacts, Environmental justice and human rights, Competition with other food uses, Health impacts of aquaculture, Labour rights, exploitation and migration, and Affordability and access each represented under 0.5% of the full evidence map.

Figure 14. *Topic-assignment frequency within the People and society theme.*

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Inputs and resources

Inputs and resources was the smallest high-level theme. Physiology, metabolism and biological responses accounted for 5,002 records (23.3%) and was more common than Growth and development (3,126; 14.5%), and Other diseases - Prevention and treatment (2,004; 9.3%). Affordability and access and Therapeutants and chemicals were represented in only 10 (<0.1%) and 9 (<0.1%) records, respectively.

Figure 15. *Topic-assignment frequency within the Inputs and resources theme.*

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Comparison with secondary research

The supplied manuscript draft compares this evidence map with a recent umbrella review of 37 systematic reviews across salmon aquaculture. Recalculation against the final canonical evidence map confirms the principal imbalance identified in the draft. Sea lice control and treatment occurs in 770 mapped records (3.6%), compared with 8 of 37 systematic reviews (21.6%) in the umbrella-review dataset. By contrast, several of the largest areas in the mapped literature have comparatively little systematic-review coverage, including Physiology, metabolism and biological responses (23.3% of mapped records), Growth and development (14.5%), and major disease, environmental and methodological topics.

The social evidence base is particularly weakly represented in the systematic-review literature described in the draft. In the final canonical map, Antimicrobial resistance and One Health accounts for 23.3% of records, Conflict and social licence for 14.5%, and Public discourse and representation for 9.3%; the supplied umbrella-review dataset contains one review on antimicrobial resistance/One Health and none in most of the other social categories. Overall, the comparison continues to indicate that the systematic-review evidence covers a narrower subset of the topics represented across the mapped literature, with particularly strong representation of sea lice and limited coverage of several major physiological, environmental, methodological, governance and social research areas.

Figure 16. *Topic-assignment frequency in the supplied umbrella review of 37 systematic reviews across salmon aquaculture. Data and figure retained from the supplied manuscript draft.*

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